Padraig Lavery

Software Engineer – Reliability & Incident Response

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Software engineer with 4+ years as the go-to engineer for production incidents on high-volume, security-sensitive banking systems running on AWS and Kubernetes. Fixes root causes rather than symptoms, makes batch systems fail safely, and turns repeat problems into runbooks or automation. Building Incident Copilot, an incident-triage service using retrieval and agentic tool-calling.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Incidents and root causes

- Resolved a live customer-facing outage caused by malformed load-balancer group naming in a Kubernetes/Istio ingress setup, and corrected the Helm deployment templates so it could not recur.
- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by loading caches once per batch instead of per file; diagnosed and mitigated further memory-exhaustion failures in ETL jobs.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so failures on the highest-volume partners no longer abort the whole run.

Preventing the next one

- Authored runbooks and SOPs (re-sending failed outbound files, re-running failed jobs) so the production services team resolves recurring issues without engineering escalation.
- Automated scheduled file processing with EventBridge, Lambda and Fargate, removing manual triggers.
- Led a cross-team initiative for automated API testing and stabilised GitHub Actions CI.

Security and upgrades

- Hardened authentication and account services by removing stack traces and SQL details from error responses and standardising input validation.
- Planned testing for AWS SDK v3 and Node 22 runtime upgrades across repositories and bank deployments, remediating critical Snyk-flagged vulnerabilities for PCI DSS.
- Mentored four new starters; run the team's retrospectives.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a cloud-cost API using Go, Docker, Kubernetes and Git.

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Takes an incident description, retrieves matching runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status.

SKILLS

- **Reliability:** incident response, root cause analysis, runbooks/SOPs, timeouts, safe retries, error isolation
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Languages:** JavaScript (Node.js), Python, C#, Go, SQL
- **Data:** PostgreSQL, MongoDB
- **Security:** Snyk, OWASP, PCI DSS remediation

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026